16 AT Commands for HTTP(S)

16.1 Overview of AT Commands for HTTP(S)

Command	Description
AT+HTTPINIT	Start HTTP service
AT+HTTPTERM	Stop HTTP Service
AT+HTTPPARA	Set HTTP Parameters value
AT+HTTPACTION	HTTP Method Action
AT+HTTPHEAD	Read the HTTP Header Information of Server Response
AT+HTTPREAD	Read the response information of HTTP Server
AT+HTTPDATA	Input HTTP Data
AT+HTTPPOSTFILE	Send HTTP Request to HTTP(S)server by File
AT+HTTPREADFILE	Receive HTTP Response Content to a file

16.2 Detailed Description of AT Commands for HTTP(S)

16.2.1 AT+HTTPINIT Start HTTP Service

AT+HTTPINIT is used to start HTTP service by activating PDP context. You must execute AT+HTTPINIT before any other HTTP related operations.

AT+HTTPINIT Start HTTP Service	
Test Command AT+HTTPINIT=?	Response **OK**
Execute Command AT+HTTPINIT	Response 1)If start HTTP service successfully: **OK** 2)If failed: **ERROR**

Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<err>	The type of error please refer to Section 16.4

Examples

```
AT+HTTPINIT
OK
```

16.2.2 AT+HTTPTERM Stop HTTP Service

AT+HTTPTERM is used to stop HTTP service.

AT+HTTPTERM Stop HTTP Service	
Test Command **AT+HTTPTERM=?**	Response **OK**
Execute Command **AT+HTTPTERM**	Response 1)If stop HTTP service successfully: **OK** 2)If failed: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Examples

```
AT+HTTPTERM
OK
```

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16.2.3 AT+HTTPPARA Set HTTP Parameters value

AT+HTTPPARA is used to set HTTP parameters value. When you want to access to a HTTP server, you should input <value> like http://'server'/'path':'tcpPort'. In addition, https://'server'/'path':'tcpPort' is used to access to a HTTPS server.

AT+HTTPPARA Set HTTP Parameters value	
Test Command **AT+HTTPPARA=?**	Response **OK**
Write Command **AT+HTTPPARA="URL",<url>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="CONNECTTO",<conn_timeout>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="RECVTO",<recv_timeout>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="CONTENT",<content_type>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="ACCEPT",<accept-type>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="SSLCFG",<sslcfg_id>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Write Command **AT+HTTPPARA="USERDATA",<user_data>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur:

	ERROR
Write Command **AT+HTTPPARA="READMODE", <readmode>**	Response 1)If parameter format is right: **OK** 2)If parameter format is not right or other errors occur: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<url>	URL of network resource.String,start with "http://" or"https://" a)http://'server' :'tcpPort' /'path'. b)https://'server' :'tcpPort' /'path'. "server" DNS domain name or IP address "path" path to a file or directory of a server "tcpPort" http default value is 80,https default value is 443.(can be omitted)
<conn_timeout>	Timeout for accessing server, Numeric type, range is 20-120s, default is 120s.
<recv_timeout>	Timeout for receiving data from server, Numeric type range is 2s-120s, default is 20s.
<content_type>	This is for HTTP "Content-Type" tag, String type, max length is 256, and default is "text/plain".
<accept-type>	This is for HTTP "Accept-type" tag, String type, max length is 256, and default is "*/*".
<sslcfg_id>	This is setting SSL context id, Numeric type, range is 0-9. Default is 0.Please refer to Chapter 19 of this document.
<user_data>	The customized HTTP header information. String type, max length is 256.
<readmode>	For HTTPREAD, Numeric type, it can be set to 0 or 1. If set to 1, you can read the response content data from the same position repeatly. The limit is that the size of HTTP server response content should be shorter than 1M.Default is 0.

NOTE

When you want to use content-type mutipart/form-data to transfer data, you should set AT+HTTPPARA="CONTENT","mutipart/form-data" .And we will construct boundary header.

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Examples

```
AT+HTTPPARA="URL","http://www.baidu.com"
OK
```

16.2.4 AT+HTTPACTION HTTP Method Action

AT+HTTPACTION is used to perform a HTTP Method. You can use HTTPACTION to send a get/post request to a HTTP/HTTPS server.

AT+HTTPACTION HTTP Method Action	
Test Command **AT+HTTPACTION=?**	Response **+HTTPACTION: (0-4)** **OK**
Write Command **AT+HTTPACTION=<method>**	Response 1)If parameter format is right: **OK** **+HTTPACTION: <method>,<statuscode>,<datalen>** 2)If parameter format is right but server connected unsuccessfully: **OK** **+HTTPACTION: <method>,<errcode>,<datalen>** 3)If parameter format is not right or other errors occur: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<method>	HTTP method specification: 0 GET 1 POST 2 HEAD 3 DELETE 4 PUT
<statuscode>	Please refer to the end of this chapter
<datalen>	The length of data received

Examples

```
AT+HTTPACTION=?
+HTTPACTION: (0-4)

OK
AT+HTTPACTION=0
OK

+HTTPACTION: 0,200,104220
```

16.2.5 AT+HTTPHEAD Read the HTTP Header Information of Server Response

AT+HTTPHEAD is used to read the HTTP header information of server response when module receives the response data from server.

AT+HTTPHEAD Read the HTTP Header Information of Server Response	
Test Command **AT+HTTPHEAD=?**	Response **OK**
Execute Command **AT+HTTPHEAD**	Response 1)If read the header information successfully: **+HTTPHEAD: <data_len>** **<data>** **OK** 2)If read failed: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<data_len>	The length of HTTP header
<data>	The header information of HTTP response

Examples

```
AT+HTTPHEAD
+HTTPHEAD: 653
```

```
HTTP/1.1 200 OK
Content-Type: text/html
Connection: keep-alive
X-Cache: MISS from PDcache-04:opinion.people.com.cn
Date: Tue, 24 Mar 2020 03:12:09 GMT
Powered-By-ChinaCache: HIT from CNC-WB-b-D24
Powered-By-ChinaCache: HIT from CNC-WV-b-D1C
ETag: W/"5b7379f5-57e9"
x-cc-via: CNC-WB-b-D24[H,1], CNC-WV-b-D1C[H,62]
d-cc-upstream: CNC-WV-b-D1C
CACHE: TCP_HIT
Vary: Accept-Encoding
Last-Modified: Wed, 15 Aug 2018 00:55:17 GMT
Expires: Tue, 24 Mar 2020 03:17:09 GMT
x-cc-req-id: f4b9e1793697d1ef2950f530aeec4519
Content-Length: 22505
Age: 0
Accept-Ranges: bytes
Server: nginx
X-Frame-Options: ALLOW-FROM .*
CC_CACHE: TCP_REFRESH_HIT
OK
```

16.2.6 AT+HTTPREAD Read the response information of HTTP Server

After sending HTTP(S)GET/POST requests, you can retrieve HTTP(S)response information from HTTP(S)server via UART/USB port by AT+HTTPREAD. When the <datalen> of "+HTTPACTION: <method>,<statuscode>,<datalen>" is not equal to 0, You can execute AT+HTTPREAD=<start_offset>,<byte_size> to read out data to port. If parameter <byte_size> is set greater than the size of data saved in buffer, all data in cache will output to port.

AT+HTTPREAD Read the response information of HTTP Server	
Test Command **AT+HTTPREAD=?**	Response **OK**
Read Command **AT+HTTPREAD?**	Response 1)If check successfully: **+HTTPREAD: LEN,<len>** **OK** 2)If failed (no more data other error): **ERROR**
Write Command	Response

AT+HTTPREAD=[<start_offset>,]<byte_size>	1)If read the response info successfully: **OK** **+HTTPREAD: <data_len>** **<data>** **+HTTPREAD: 0** If <byte_size> is bigger than the data size received, module will only return actual data size. 2)If read failed: **ERROR**
Parameter Saving Mode	-
Max Response Time	120000ms
Reference	-

Defined Values

<start_offset>	The start position of reading
<byte_size>	The length of data to read
<data_len>	The actual length of read data
<data>	Response content from HTTP server
<len>	Total size of data saved in buffer.

Examples

```
AT+HTTPREAD?
+HTTPREAD: LEN,22505

OK
AT+HTTPREAD=0,500
OK

+HTTPREAD: 500
\0\0\0\0\0\0\0\0\0\0\0\0\0\0<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN"
"http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<meta http-equiv="content-type" content="text/html;charset=GB2312"/>
<meta http-equiv="Content-Language" content="utf-8" />
<meta content="all" name="robots" />
<title>人民日报钟声:牢记历史是为了更好开创未来--观点--人民网 </title>
<meta name="keywords" content="" />
<meta name="description" content="      日方应在正确对待历史?
+HTTPREAD: 0
```

NOTE

The response content received from server will be saved in cache, and would not be cleaned up by AT+HTTPREAD.
Due to the max size of protocol stack is 64K bytes(the CAT4 module is 10K bytes), when the total size of the data from server is bigger than that and 'READMODE' is 0, you should read the data quickly, or you will fail to read it.

16.2.7 AT+HTTPDATA Input HTTP Data

You can use AT+HTTPDATA to input data to post when you send a HTTP/HTTPS POST request.

AT+HTTPDATA Input HTTP Data	
Test Command **AT+HTTPDATA=?**	Response **OK**
Write Command **AT+HTTPDATA=<size>,<time>**	Response 1)if parameter format is right: **DOWNLOAD** **<input data here>** When the total size of the inputted data reaches <size>, TA will report the following code. Otherwise, the serial port will be blocked. **OK** 2)If parameter format is wrong or other errors occur: **ERROR**
Parameter Saving Mode	
Max Response Time	
Reference	

Defined Values

<size>	Size in bytes of the data to post. range is 1- 153600 (bytes)
<time>	Maximum time in seconds to input data.range is 10-65535

Examples

```
AT+HTTPDATA=18,1000
DOWNLOAD
Message=helloworld
OK
```

16.2.8 AT+HTTPPOSTFILE Send HTTP Request to HTTP(S)server by File

You also can send HTTP request in a file via AT+HTTPPOSTFILE command. The URL must be set by AT+HTTPPARA before executing AT+HTTPPOSTFILE command. The parameter <path> can be used to set the file directory. When modem has received response from HTTP server, it will report the following URC:
+HTTPPOSTFILE: <statuscode>,<datalen>

AT+HTTPPOSTFILE Send HTTP Request to HTTP(S)server by File	
Test Command **AT+HTTPPOSTFILE=?**	Response **+HTTPPOSTFILE: <filename>[,(1-2)[,(0-4)[,(0-1)]]]** **OK**
Write Command **AT+HTTPPOSTFILE=<filename>[,<path>[,<method>[,<send_header>]]]**	Response 1)if parameter format is right and server connected successfully: a)if parameter <method> is valid: **OK** **+HTTPPOSTFILE: <method>,<statuscode>,<datalen>** b)if parameter <method> is ignored: **OK** **+HTTPPOSTFILE: <statuscode>,<datalen>** 2)if parameter format is right but server connected unsuccessfully: a)if parameter <method> is valid: **OK** **+HTTPPOSTFILE: <method>,<errcode>,0** b)if parameter <method> is ignored: **OK** **+HTTPPOSTFILE: <errcode>,0** 3)if parameter format is not right or any other error occurs: **ERROR**
Parameter Saving Mode	
Max Response Time	
Reference	

Defined Values

<filename>	String type, filename, the max length is 112.unit:byte.
<path>	The directory where the sent file saved. Numeric type, range is 1-2 1 C:/ (local storage) 2 D:/(sd card)
<method>	HTTP method specification: 0 GET 1 POST 2 HEAD 3 DELETE 4 PUT If this value is not provided, it is same to the value described in the post file.
<send_header>	Send file as HTTP header and Body or Only as Body. Numeric type, the range is 0-1, the default is 0. 0 Send file as Body 1 Send file as HTTP header and body
<statuscode>	Please refer to the end of this chapter
<datalen>	The length of data received

Examples

```
AT+HTTPPOSTFILE=?
+HTTPPOSTFILE: <filename>[,(1-2)[,(0-4)[,(0-1)]]]

OK
AT+HTTPPOSTFILE="getbaidu.txt",1
OK

+HTTPPOSTFILE: 200,14615
AT+HTTPPOSTFILE="getbaidu.txt",1,1,1
OK

+HTTPPOSTFILE: 1,200,14615
```

16.2.9 AT+HTTPREADFILE Receive HTTP Response Content to a file

After execute AT+HTTPACTION/AT+HTTPOSTFILE command. You can receive the HTTP server response

content to a file via AT+HTTPREADFILE.
Before AT+HTTPREADFILE executed, "+HTTPACTION: <method>,<statuscode>,<datalen>" or "+HTTPPOSTFILE: <statuscode>,<datalen>" must be received. The parameter <path> can be used to set the directory where to save the file. If omit parameter <path>, the file will be save to local storage.

AT+HTTPREADFILE Receive HTTP Response Content to a File	
Test Command **AT+HTTPREADFILE=?**	Response **+HTTPREADFILE: <filename>[,(1-2)]** **OK**
Write Command **AT+HTTPREADFILE=<filename>[,<path>]**	Response 1)if parameter format is right: **OK** **+HTTPREADFILE: <errcode>** 2)if failed: **OK** **+HTTPREADFILE: <errcode>** 3)if parameter format is not right or any other error occurs: **ERROR**
Parameter Saving Mode	
Max Response Time	
Reference	

Defined Values

<filename>	String type, filename, the max length is 112.unit:byte.
<path>	The directory where the read file saved. Numeric type, range is 1-2. 1 C:/(local storage) 2 D:/(sd card)

Examples

```
AT+HTTPREADFILE=?
+HTTPREADFILE: <filename>[,(1-2)]

OK
AT+HTTPREADFILE="readbaidu.dat"
OK

+HTTPREADFILE: 0
```

16.3 Command Result Codes

16.3.1 Description of <statuscode>

<statuscode>	Description
100	Continue
101	Switching Protocols
200	OK
201	Created
202	Accepted
203	Non-Authoritative Information
204	No Content
205	Reset Content
206	Partial Content
300	Multiple Choices
301	Moved Permanently
302	Found
303	See Other
304	Not Modified
305	Use Proxy
307	Temporary Redirect
400	Bad Request
401	Unauthorized
402	Payment Required
403	Forbidden
404	Not Found
405	Method Not Allowed
406	Not Acceptable
407	Proxy Authentication Required
408	Request Timeout
409	Conflict
410	Gone
411	Length Required
412	Precondition Failed
413	Request Entity Too Large
414	Request-URI Too Large

415	Unsupported Media Type
416	Requested range not satisfiable
417	Expectation Failed
500	Internal Server Error
501	Not Implemented
502	Bad Gateway
503	Service Unavailable
504	Gateway timeout
505	HTTP Version not supported
600	Not HTTP PDU
601	Network Error
602	No memory
603	DNS Error
604	Stack Busy

16.3.2 Description of <errcode>

<errcode>	Meaning
0	Success
701	Alert state
702	Unknown error
703	Busy
704	Connection closed error
705	Timeout
706	Receive/send socket data failed
707	File not exists or other memory error
708	Invalid parameter
709	Network error
710	start a new ssl session failed
711	Wrong state
712	Failed to create socket
713	Get DNS failed
714	Connect socket failed
715	Handshake failed
716	Close socket failed
717	No network error
718	Send data timeout
719	CA missed

16.4 Unsolicited Result Codes

URC	Description
+HTTP_PEER_CLOSED	It's a notification message. While received, it means the connection has been closed by server.
+HTTP_NONET_EVENT	It's a notification message. While received, it means now the network is unavailable.

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